3. A fair six-sided black die has faces numbered 1, 2, 2, 3, 3 and 4

The random variable B represents the score when the black die is rolled.

(a) Write down the value of $\mathrm{E}(B)$

(1)

A white die has 6 faces numbered 1, 1, 2, 4, 5 and c where $c > 5$
The discrete random variable W represents the score when the white die is rolled and has probability distribution given by

w	1	2	4	5	c
$\mathrm{P}(W = w)$	$a + b$	a	0.3	a	b

Greg and Nilaya play a game with these dice.

Greg throws the black die and Nilaya throws the white die. Greg wins the game if he scores at least two more than Nilaya, otherwise Greg loses.

The probability of Greg winning the game is $\dfrac{1}{6}$

(b) Find the value of a and the value of b
Show your working clearly.

(5)

The random variable $X = 2W - 5$

Given that $\mathrm{E}(X) = 2.6$

(c) find the exact value of $\mathrm{Var}(X)$

(6)

4. The discrete random variable X has probability distribution

x	-3	-1	1	2	4
$P(X = x)$	q	$\frac{7}{30}$	$\frac{7}{30}$	q	r

where q and r are probabilities.

(a) Write down, in terms of q, $P(X \leqslant 0)$ **(1)**

(b) Show that $E(X^2) = \frac{7}{15} + 13q + 16r$ **(2)**

Given that $E(X^3) = E(X^2) + E(6X)$

(c) find the value of q and the value of r **(7)**

(d) Hence find $P(X^3 > X^2 + 6X)$ **(4)**

1. A plumbing company receives call-outs during the working day at an average rate of 2.4 per hour.

(a) Find the probability that the company receives exactly 7 call-outs in a randomly selected 3-hour period of a working day.

(2)

The company has enough staff to respond to 28 call-outs in an 8-hour working day.

(b) Show that the probability that the company receives more than 28 call-outs in a randomly selected 8-hour working day is 0.022 to 3 decimal places.

(2)

In a random sample of 100 working days each of 8 hours,

(c) (i) find the expected number of days that the company receives more than 28 call-outs,

(1)

(ii) find the standard deviation of the number of days that the company receives more than 28 call-outs,

(2)

(iii) use a Poisson approximation to estimate the probability that the company receives more than 28 call-outs on at least 6 of these days.

(3)

3. The probability distribution of the discrete random variable X is

$$\mathrm{P}(X = x) = \begin{cases} \dfrac{k}{x} & \text{for } x = 1, 2 \text{ and } 3 \\ \dfrac{m}{2x} & \text{for } x = 6 \text{ and } 9 \\ 0 & \text{otherwise} \end{cases}$$

where k and m are positive constants.

Given that $\mathrm{E}(X) = 3.8$, find $\mathrm{Var}(X)$

(7)

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3. The discrete random variable X has probability distribution

x	-3	-2	-1	0	2	5
$\mathrm{P}(X = x)$	0.3	0.15	0.1	0.15	0.1	0.2

(a) Find $\mathrm{E}(X)$

(1)

Given that $\mathrm{Var}(X) = 8.79$

(b) find $\mathrm{E}(X^2)$

(2)

The discrete random variable Y has probability distribution

y	-2	-1	0	1	2
$\mathrm{P}(Y = y)$	$3a$	a	b	a	c

where a, b and c are constants.

For the random variable Y

$$\mathrm{P}(Y \leqslant 0) = 0.75 \quad \text{and} \quad \mathrm{E}(Y^2 + 3) = 5$$

(c) Find the value of a, the value of b and the value of c

(5)

The random variable $W = Y - X$ where Y and X are independent.

The random variable $T = 3W - 8$

(d) Calculate $\mathrm{P}(W > T)$

(4)

4. The discrete random variable X has the following probability distribution

x	0	2	3	6
$P(X = x)$	p	0.25	q	0.4

(a) Find in terms of q

(i) $E(X)$

(ii) $E(X^2)$

(2)

Given that $\text{Var}(X) = 3.66$

(b) show that $q = 0.3$

(3)

In a game, the score is given by the discrete random variable X

Given that games are independent,

(c) calculate the probability that after the 4th game has been played, the total score is exactly 20

(3)

A round consists of 4 games plus 2 bonus games. The bonus games are only played if after the 4th game has been played the total score is exactly 20

A prize of £10 is awarded if 6 games are played in a round **and** the total score for the round is at least 27

Bobby plays 3 rounds.

(d) Find the probability that Bobby wins at least £10

(6)

1. The discrete random variable X has the following distribution

x	0	1	2	3	4
$P(X = x)$	r	k	$\frac{k}{2}$	$\frac{k}{3}$	$\frac{k}{4}$

where r and k are positive constants.

The standard deviation of X equals the mean of X

Find the exact value of r

(6)

3. A machine produces cloth. Faults occur randomly in the cloth at a rate of 0.4 per square metre.

The machine is used to produce tablecloths, each of area A square metres. One of these tablecloths is taken at random.

The probability that this tablecloth has no faults is 0.0907

(a) Find the value of A

(3)

The tablecloths are sold in packets of 20

A randomly selected packet is taken.

(b) Find the probability that more than 1 of the tablecloths in this packet has no faults.

(3)

A hotel places an order for 100 tablecloths each of area A square metres.

The random variable X represents the number of these tablecloths that have no faults.

(c) Find

(i) $\mathrm{E}(X)$

(ii) $\mathrm{Var}(X)$

(3)

(d) Use a Poisson approximation to estimate $\mathrm{P}(X = 10)$

(2)

It is claimed that a new machine produces cloth with a rate of faults that is less than 0.4 per square metre.

A piece of cloth produced by this new machine is taken at random.
The piece of cloth has area 30 square metres and is found to have 6 faults.

(e) Stating your hypotheses clearly, use a suitable test to assess the claim made for the new machine. Use a 5% level of significance.

(4)

(f) Write down the p-value for the test used in part (e).

(1)

3. The discrete random variable X has probability distribution,

x	-1	0	1	3	7
$\mathrm{P}(X = x)$	p	r	p	0.3	r

where p and r are probabilities.

Given that $\mathrm{E}(X) = 1.95$

find the exact value of $\mathrm{E}\left(\sqrt{X+1}\right)$ giving your answer in the form $a + b\sqrt{2}$ where a and b are rational.

(6)

2. The discrete random variable X has probability distribution given by

x	-1	0	1	2	3
$P(X = x)$	c	a	a	b	c

The random variable $Y = 2 - 5X$

Given that $E(Y) = -4$ and $P(Y \geqslant -3) = 0.45$

(a) find the probability distribution of X.

(7)

Given also that $E(Y^2) = 75$

(b) find the exact value of $\text{Var}(X)$

(2)

(c) Find $P(Y > X)$

(2)

3. Two car hire companies hire cars independently of each other.

Car Hire A hires cars at a rate of 2.6 cars per hour.

Car Hire B hires cars at a rate of 1.2 cars per hour.

(a) In a 1 hour period, find the probability that each company hires exactly 2 cars.
(2)

(b) In a 1 hour period, find the probability that the total number of cars hired by the two companies is 3
(2)

(c) In a 2 hour period, find the probability that the total number of cars hired by the two companies is less than 9
(2)

On average, 1 in 250 new cars produced at a factory has a defect.

In a random sample of 600 new cars produced at the factory,

(d) (i) find the mean of the number of cars with a defect,

(ii) find the variance of the number of cars with a defect.
(2)

(e) (i) Use a Poisson approximation to find the probability that no more than 4 of the cars in the sample have a defect.

(ii) Give a reason to support the use of a Poisson approximation.
(2)